

Exp :5 Design a lexical Analyzer to find the number of whitespaces and newline characters using C.

```
#include <stdio.h>

int main() {

    char str[100];

    int words = 0, lines = 0, characters = 0;

    int in_word = 0;

    printf("Enter text (up to 100 characters, use ~ to end):\n");

    scanf("%[^\n]", str);

    for (int i = 0; str[i] != '\0'; i++) {

        characters++;

        if (str[i] == '\n') {

            lines++;

        }

        if (str[i] == ' ' || str[i] == '\t' || str[i] == '\n') {

            in_word = 0;

        } else if (in_word == 0) {

            in_word = 1;

            words++;

        }

    }

    if (characters > 0 && lines == 0) {

        lines = 1;

    }

    printf("Total number of words: %d\n", words);

    printf("Total number of lines: %d\n", lines);

    printf("Total number of characters: %d\n", characters);

}
```

```
    return 0;
}
```

Output:

```
Enter text (up to 100 characters, use ~ to end):
void main()
int a;
int b;
~
Total number of words: 6
Total number of lines: 3
Total number of characters: 26

...Program finished with exit code 0
Press ENTER to exit console.
```